

```
from google.colab import files
uploaded = files.upload()
```

 test_set.zip

test_set.zip(application/x-zip-compressed) - 46210370 bytes, last modified: 7/17/2026 - 100% done
Saving test_set.zip to test_set (1).zip

```
import zipfile

with zipfile.ZipFile('/content/test_set.zip', 'r') as zip_ref:
    zip_ref.extractall('/content')
```

```
import os

print(os.listdir('/content'))
```

```
print(os.listdir('/content/test_set'))
```

```
import os
import cv2
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report
from sklearn.utils import shuffle
```

```
import os

print(os.listdir('/content'))
print(os.listdir('/content/test_set'))
```

```
print(os.listdir('/content/test_set'))
```

```
import os
import cv2
import numpy as np

data = []
labels = []

IMG_SIZE = 64

DATADIR = "/content/test_set"
categories = ["cats", "dogs"]

for category in categories:

    path = os.path.join(DATADIR, category)

    label = categories.index(category)

    count = 0
```

```
for img in os.listdir(path):  
    try:  
        img_path = os.path.join(path, img)  
  
        image = cv2.imread(img_path)  
  
        if image is not None:  
  
            image = cv2.resize(image, (64,64))  
  
            data.append(image)  
            labels.append(label)  
  
            count += 1  
  
            if count == 500:  
                break  
  
    except:  
        pass  
  
print("Images Loaded :", len(data))
```

```
X = np.array(data)  
y = np.array(labels)  
  
print(X.shape)  
print(np.unique(y))
```

```
from sklearn.utils import shuffle  
  
X, y = shuffle(X, y, random_state=42)  
  
X = X.reshape(len(X), -1)  
  
X = X / 255.0
```

```
from sklearn.model_selection import train_test_split  
  
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.2,  
    random_state=42,  
    stratify=y  
)
```

```
print(np.unique(y_train))  
print(np.unique(y_test))
```

```
[0 1]  
[0 1]
```

```
from sklearn.svm import SVC  
  
model = SVC(kernel='linear')
```

```
model.fit(X_train, y_train)
```

▼ SVC ⓘ ?
SVC(kernel='linear')

```
y_pred = model.predict(X_test)
```

```
from sklearn.metrics import accuracy_score, classification_report  
  
print("Accuracy :", accuracy_score(y_test, y_pred))  
  
print(classification_report(y_test, y_pred))
```

```
from sklearn.metrics import confusion_matrix  
import matplotlib.pyplot as plt  
  
cm = confusion_matrix(y_test, y_pred)  
  
plt.figure(figsize=(6,5))  
  
plt.imshow(cm, cmap='Blues')  
plt.colorbar()  
  
plt.xticks([0,1], ['Cat','Dog'])  
plt.yticks([0,1], ['Cat','Dog'])  
  
plt.xlabel('Predicted Label')  
plt.ylabel('True Label')  
plt.title('Confusion Matrix')  
  
for i in range(2):  
    for j in range(2):  
        plt.text(j, i, cm[i, j],  
                  ha='center',  
                  va='center',  
                  fontsize=14)  
  
plt.show()
```

